

Along Equal Lines

With the tin-can versions gone, we're wondering what the new humanoid-type robots will be up to

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Researchers are falling over themselves to re-enact The Creation, and people are already accepting robots as friends, companions and lovers. In Japan, household robots number in the millions. Most robots today are things like vacuum cleaners, and the rest are doing some cute, some interesting, and some boring things like carrying on micro-conversations ("What is your name?"), climbing stairs, saying "Whoa!" when pushed, playing football, eating sushi, and serving as receptionists. (Spot which one we made up.)

The mindset towards robots has changed in a surprisingly short span of time. A couple of decades ago, they were seen as curious toys, or as miracles, or as dangerous entities that would threaten our domination of the planet—in short, much ado was being made about them. Today, tell someone "Fitzy" won the annual table tennis championship for robots, and he'll tell you who won the rock-climbing championship.

We're amazed at the intelligent things these once-toys can do. What will they be able to do a decade from now? Thirty years? Will it be like in *Bicentennial Man*, where they're almost part of society, with the same status we accord to ourselves? Will they be classified as a species? Will they have noisy sex? Just how intelligent can they really get? Etc. etc. etc. Of course we don't know any of the answers, but it won't hurt to take a romp through the questions.

Eye To Eye

Pong, the Sympathetic Robot, was developed several years ago at IBM's Almaden Research facility. It is a moving head that "sits on a table and looks at you with two ping-pong ball-like eyes; once it makes eye contact, it smiles or frowns based on its interpretation of your mood.

"I found it fascinating and creepy at the same time. There is something deeply unnerving about a machine looking at you. If the researchers replaced the ping ball eyes with realistic glass eyes, I'm sure it would send shivers down your spine.

"The idea that a machine can approach humans may be deeply repugnant. Perhaps we do not want to develop thinking machines that can look you in the eye."

